UV-C Disinfection Robotic Car

***Abstract*—In response to the urgent demand for cost-effective and environmentally conscious disinfection solutions, this project presents an innovative Ecofriendly UVC Disinfecting Robot. This endeavor capitalizes on easily obtainable components, including the Arduino Uno, L298n motor driver, sonar sensor, and IR sensors, to construct an economical, automated disinfection system. The amalgamation of these components empowers the robot with autonomous mobility, obstacle detection capabilities, and precise UVC disinfection functionality.**

The primary objective of the robot revolves around the proficient deployment of UVC (Ultraviolet-C) light for thorough surface disinfection, offering a sustainable substitute for conventional chemical-based disinfection methods. UVC light is renowned for its germicidal properties, rendering it a secure and exceedingly efficient tool for eradicating a diverse spectrum of harmful microorganisms, encompassing bacteria and viruses.

***Index Terms*—UVC, Disinfection robot, Eco-friendly**

I. Introduction

In response to the growing demand for affordable and eco-friendly disinfection methods, we present the "UVC Disinfection Robot." This paper delves into the design, assembly, and operational aspects of our robot, emphasizing its utilization of basic, cost-effective components. While not pushing the boundaries of technological sophistication, our robot offers a practical solution for addressing microbial threats in various environments.

Our UVC Disinfection Robot is intentionally constructed using readily available components to keep costs low. It employs a sonar sensor for obstacle avoidance and an IR sensor for basic room navigation. These simple sensors facilitate the robot's movement within its environment, albeit with some limitations.

At the core of our robot's disinfection capabilities lies a series of UVC LEDs mounted under its base chassis, positioned to disinfect the area beneath it. Ultraviolet-C (UVC) light is known for its germicidal properties, effectively eliminating a range of harmful microorganisms, including bacteria and viruses. UVC light accomplishes this by disrupting the genetic material of these microorganisms, primarily DNA and RNA.

When exposed to UVC light at a specific wavelength (around 254 nanometers), the photons in the light are absorbed by the genetic material of the microorganisms. This absorption of energy leads to a process known as photodimerization, where adjacent thymine or cytosine bases on the DNA or RNA strands bind together, forming abnormal structures. These abnormal structures inhibit the microorganism's ability to replicate and carry out essential cellular functions, rendering them inactive and incapable of causing infection.

By harnessing the germicidal power of UVC light, our robot provides an eco-friendly alternative to chemical-based disinfection methods, promoting healthier living and working spaces. This paper offers a comprehensive overview of the UVC Disinfection Robot's design and assembly, emphasizing its reliance on cost-effective components and elucidating the science behind UVC disinfection. We also present experimental results demonstrating the robot's effectiveness in surface disinfection, acknowledging the trade-off between sophistication and cost-effectiveness in the design of such robots.

The following are the novelty of our project compared to existing work:

1) Implements an efficient PID control mechanism using the FastPID library, applied at regular intervals via an ISR (Interrupt Service Routine).
2) Operates in two distinct modes: Mapping Mode (saves data to Arduino UNO's EEPROM) and Normal Mode (uses saved data for navigation while dynamically avoiding obstacles such as humans).
3) Achieves a highly cost-effective design by eliminating the need for additional memory cards or complex hardware components.

We have made the source code of our project publicly available for further research [4], along with a technical explanation of the programmed algorithm.

II. Literature Review

The development of disinfection robots, particularly those utilizing UVC (Ultraviolet-C) light, has gained significant attention in recent years due to the pressing need for effective and sustainable disinfection solutions. In this literature review, we explore the existing body of research related to UVC disinfection and robotic systems, with a focus on their integration into cost-effective and environmentally friendly solutions.

UVC light is renowned for its germicidal properties, effectively deactivating microorganisms by disrupting their genetic material. Numerous studies have demonstrated the effectiveness of UVC light in disinfecting surfaces and air, making it a promising technology for pathogen control.

Robotic systems have emerged as versatile tools in various industries due to their efficiency in automating tasks. In the context of disinfection, robots equipped with sensors for navigation and mapping have been developed to ensure consistent and precise disinfection coverage.

Iong Chio [1] has explored the integration of UVC technology into disinfection robots, enhancing the efficiency of disinfection processes. These robots are equipped with UVC emitters, mounted on mobile platforms, to deliver targeted disinfection in various environments.

In response to the need for affordable solutions, there is growing interest in developing cost-effective disinfection robots. These robots utilize readily available components and open-source platforms to reduce production costs while maintaining effectiveness.

An emerging trend in disinfection robots is their ability to serve multiple functions. Some robots can be equipped with additional modules, such as vacuum suction, to transform them into versatile cleaning and maintenance tools, enhancing their practicality.

Recent research conducted by M. Sedaghat [2] has shown that UVC multi-purpose robots can achieve a remarkable 80 percent disinfection of microbes within just 20 minutes. This finding underscores the practicality and efficiency of UVC disinfection technology when integrated into robotic systems.

III. Methodology

The UVC Disinfection Robot is designed to navigate through its environment while efficiently disinfecting surfaces using UVC (Ultraviolet-C) light. This section outlines the key components and their functions in the robot's working mechanism.

A. Sonar Sensor for Obstacle Detection

The robot is equipped with a sonar sensor, positioned at its front, which constantly emits ultrasonic waves. When these waves encounter an obstacle in the robot's path, they bounce back and are detected by the sensor. Upon detecting an obstacle within a certain proximity, the robot's control system triggers a stop command.

B. IR Sensor for Directional Navigation

To navigate around obstacles, the robot relies on an IR (Infrared) sensor system. This IR sensor system is strategically placed on the robot's sides, allowing it to detect the absence of obstacles in either the left or right direction. Depending on the IR sensor's signal, the robot makes a decision to move either left or right to circumvent the obstacle. The robot continuously monitors the IR sensor signals to ensure safe and obstacle-free navigation.

C. Wheel Encoders with PID Controller for Straight Path Travel

To maintain a straight path during its movement, the robot utilizes optical wheel encoders. These encoders are attached to the robot's wheels and provide feedback about the wheel's rotational motion. The control system constantly compares the signals from both wheel encoders to ensure that both wheels are turning at the same rate. To modulate the error signal and precisely control the robot's movement, a PID (Proportional-Integral-Derivative) feedback loop is employed. The PID controller adjusts the wheel speeds based on the error signal, ensuring that the robot travels in a straight line.

D. UVC Disinfection Functionality

Simultaneously with its navigation, the robot emits UVC light from the mounted UVC LEDs at its base. This UVC light irradiates and disinfects the surface area directly beneath the robot as it moves. UVC light is known for its germicidal properties, effectively deactivating a wide range of harmful microorganisms, including bacteria and viruses. To achieve a lethal dose of $3.5mJ/cm^2$, we calculated the required exposure time based on the UVC light intensity emitted by the LEDs. We set the robot's speed and adjusted its movement to ensure that the UVC light effectively irradiates each surface area for the specified duration.

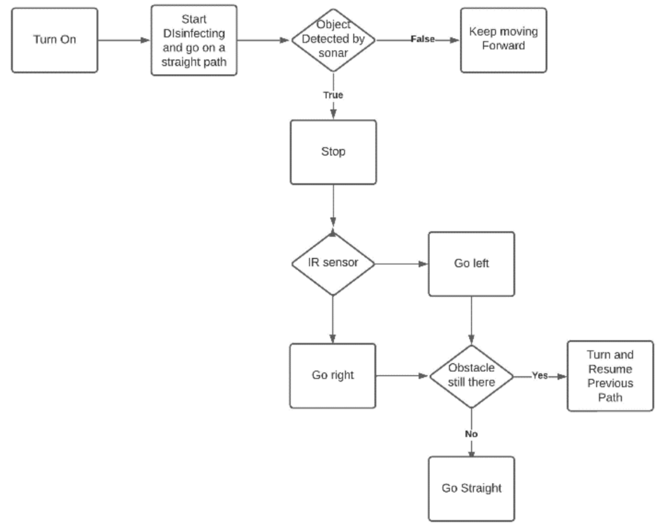

Fig. 1. Algorithm for our robotic car's movement

In essence, the UVC Disinfection Robot employs a multi-sensor navigation system, integrating a sonar sensor, IR sensors, wheel encoders, and a PID controller to navigate autonomously while avoiding obstacles, maintaining a straight path, and accurately modulating its movements. Concurrently, it administers UVC disinfection to surfaces, offering a comprehensive and efficient solution for safe and sustainable disinfection.

E. Hardware Implementation

The hardware components used in this project are:

1) **UV-C LEDs**: Emit germicidal UV light for disinfection.
2) **Arduino UNO**: Controls robot functions and sensors.
3) **IR Sensors**: Detect obstacles for navigation.
4) **Sonar Sensor**: Detects front obstacles to halt the robot.
5) **DC Gear Motors**: Drive robot movement.
6) **L298n Motor Driver**: Controls motor functions.
7) **Wheels**: Provide traction for mobility.
8) **Plastic Chassis and Cover**: Holds and supports the whole system

F. Result analysis

One notable achievement is the robot's operational efficiency. Through our tests, we found that the robot can disinfect an area of up to 120 square meters on a single charge. This demonstrates its potential to provide thorough disinfection coverage in different settings, promoting a safer environment.

In this phase, we primarily focused on the robot's hardware, sensors, and mobility. Our next steps involve conducting

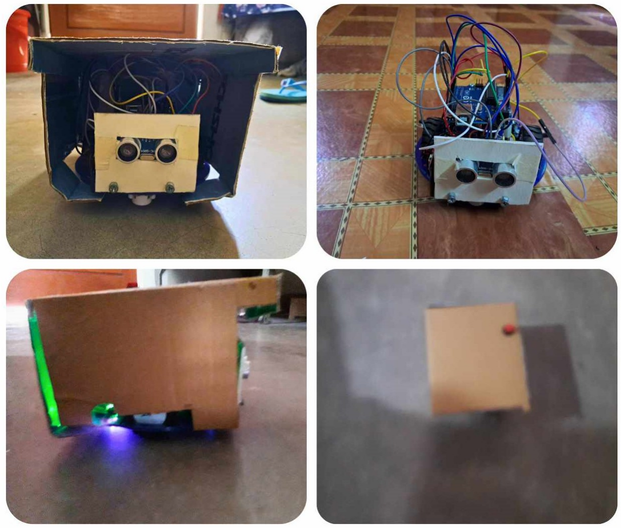

Fig. 2. Final Prototype

thorough disinfection tests to validate its performance and ensure it effectively deactivates harmful microorganisms.

In summary, the UVC Disinfection Robot is a promising development in autonomous disinfection technology. Although we haven't conducted formal tests yet, we're confident in its capabilities based on its design and efficiency. Further testing will provide concrete evidence of its effectiveness in creating safer and healthier environments.

G. Environment Friendly Aspects

Our project embodies several environmentally friendly features and practices that align with the principles of sustainability and responsible resource management:

- **Use of Recyclable Materials**: The body of the UVC Disinfection Robot is constructed using recyclable materials. This design choice minimizes the environmental footprint by ensuring that the robot's components can be repurposed or recycled at the end of their lifecycle.
- **Energy-Efficient Operation**: Our robot operates on an energy-efficient basis by utilizing low-power components. The inclusion of optimization algorithms for movement ensures that the robot consumes minimal energy during its tasks.
- **Modular Design for Sustainability**: The modular design of our robot is a key feature that enhances sustainability. Instead of discarding the entire robot when certain parts become obsolete or damaged, our design allows for the replacement or upgrading of individual components. This approach prolongs the robot's lifespan and reduces electronic waste, promoting a more environmentally responsible and cost-effective solution.

These environmental sustainability aspects reflect our commitment to minimizing the ecological footprint of our project while maximizing its efficiency and adaptability. By incorporating these practices, we contribute to a greener and more sustainable future while addressing critical disinfection needs in various environments.

H. Cost Analysis

Component	Quantity	Cost (BDT)
Arduino UNO	1	1100
UV-C LEDs	10	200
Rechargeable Li-ion Battery	2	200
Battery Charger	1	100
IR Sensors	2	200
Sonar Sensor	1	100
DC Gear Motor	2	600
Chassis	1	300
L298n Motor Driver	1	200
Wheels, Breadboard, etc.	-	500
Total Cost		**3500**

Fig. 3. Cost Anaysis

IV. Comparison with Existing Work

Our UVC Disinfection Robot stands out in several key aspects when compared to existing industrial disinfection robots and traditional mopping robots.

A. Cost-Effectiveness and Versatility

Designed with cost-effectiveness in mind, our robot is well-suited for home, office, and hospital usage. It leverages readily available components, making it affordable and accessible for a wide range of users. Many existing industrial disinfection robots are tailored for large-scale applications and come with a hefty price tag. They may be financially out of reach for smaller-scale or non-industrial users.

B. Adaptability as a Mopper

One of the standout features of our robot is its adaptability as a mopper. By equipping it with appropriate brushes or attachments, it can transform into a versatile cleaning and maintenance tool. Traditional mopping methods are labor-intensive and often require separate equipment for both disinfection and cleaning. Our robot combines both functions, streamlining the process.

C. Mobility and Coverage

Our robot's mobility (due to compact size) and obstacle detection capabilities enable it to reach and disinfect hard-to-reach areas efficiently. Some existing industrial disinfection robots may be bulkier and less agile, limiting their ability to access tight spaces.

V. Future Prospects

There are certain areas for improvement for our robot. Here are some key areas we're considering for our UVC disinfection robot.

A. *Autonomous Navigation Improvement*

We believe that investing in advanced algorithms and cutting-edge navigation technologies is crucial. This will empower our robot to adapt to dynamic environments, navigate complex obstacles, and optimize its disinfection routes autonomously.

B. *Remote Monitoring and Control*

To scale up our operations efficiently, we're exploring the development of a remote monitoring and control system. This would allow us to oversee multiple robots simultaneously, track their progress in real-time, and intervene when necessary, particularly in larger deployments.

C. *Data Analytics and Reporting*

Data-driven insights are invaluable. Implementing data analytics capabilities will enable us to analyze the disinfection data collected by our robot. This analysis will provide us with a comprehensive understanding of disinfection coverage, patterns, and effectiveness. Our goal is to create user-friendly reports that can benefit facility managers and health authorities.

REFERENCES

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[4] Github link: https://github.com/SubhanBihan/UVC-Car